

HAT-P-3b

R-1168

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-3_210528 · created 2026-07-31T03:05:26+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459362.8965 +/- 0.0026 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.114 +/- 0.014
Transit depth (Rp/Rs)^2	1.29 +/- 0.32 [%]
Orbital Inclination (inc)	88.62 +/- 0.79
Airmass coefficient 1 (a1)	1.04 +/- 0.016
Airmass coefficient 2 (a2)	-0.0184 +/- 0.0027
Scatter in the residuals of the lightcurve fit is	1.49 %
Best Comparison Star	#2 - [118, 66]
Optimal Aperture	3.78
Optimal Annulus	13.35
Transit Duration (day)	0.0957 +/- 0.0035

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.114 ± 0.014 vs 0.1063 (deviation 0.37σ)
Duration fit vs published	0.0957 d vs 0.0867 d (deviation 1.75σ)
Mid-time O–C	2.3 min
Depth SNR	5.5
Residual scatter	1.49 %

Light curve

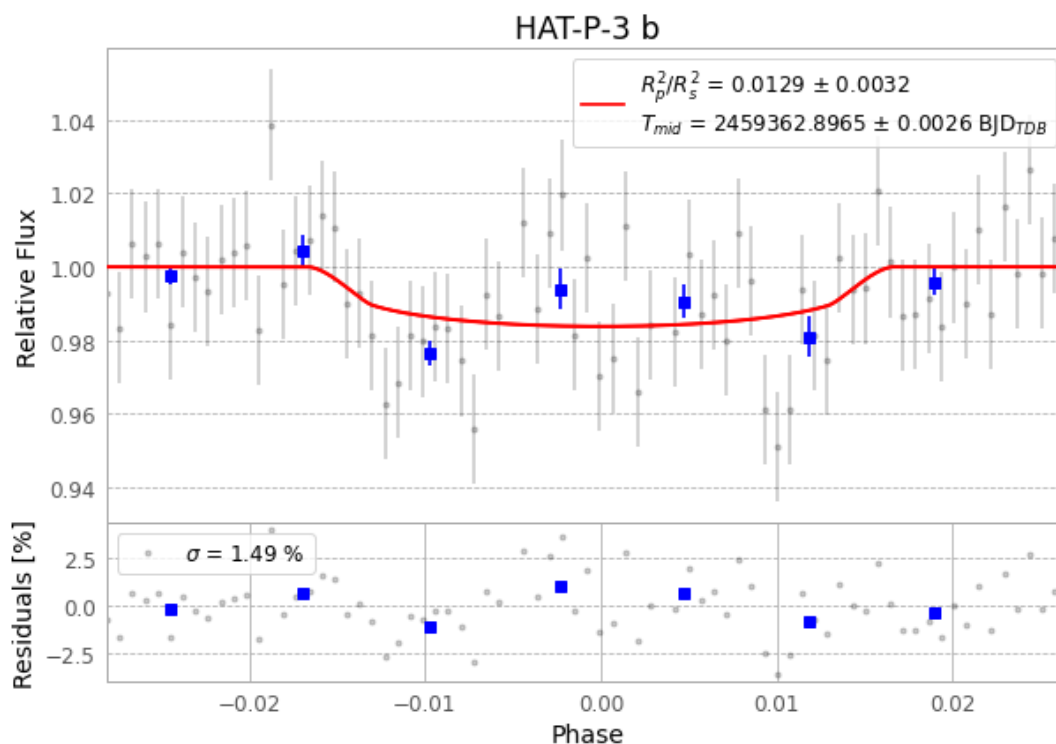


Figure 1 — FinalLightCurve_HAT-P-3 b_2021-05-28.png (SHA-256 b903dc466d3df259...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [402, 287], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 aa437dc9fe0cca45...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

77 FITS frames (51 MB), HATP-3210528073012.FITS → HATP-3210528111812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-3-210528.log (7343c610febc...), FinalLightCurve_HAT-P-3 b_2021-05-28.png (b903dc466d3d...), FinalLightCurve_HAT-P-3 b_2021-05-28.csv (c8284dde926a...)

Compute resources

Wall time 105.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 03:05Z.